AP CALCULUS AB

Unit 1 • Lesson 1.14 notes

Lesson 1.14 • Connecting Infinite Limits and Vertical Asymptotes

Lesson notes • Learning goals, key ideas, worked examples, common mistakes, quick check

Learning goals

- Describe unbounded behavior with infinite limits.
- Connect infinite limits to vertical asymptotes, and determine signs.

Vocabulary. infinite limit, vertical asymptote, unbounded.

Key idea

If $\lim_{x\to c^-} f(x)$ or $\lim_{x\to c^+} f(x)$ is $+\infty$ or $-\infty$, the line $x = c$ is a vertical asymptote. Writing $\lim f(x) = \infty$ describes *how* the limit fails to exist.

1. Determining the sign

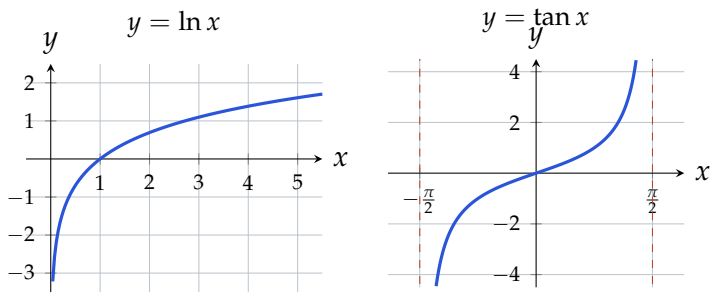
For $\frac{k}{0}$ forms, find the sign of the numerator and of each denominator factor just left and just right of c.

Example 1

$f(x) = \dfrac{x-4}{x(x-2)}$ near $x = 2$.

Solution. The numerator is near -2 and x is near 2. For $x < 2$, $x - 2 < 0$: denominator negative, so $f > 0$: $\lim_{x\to 2^-} f(x) = +\infty$. For $x > 2$: $\lim_{x\to 2^+} f(x) = -\infty$.

Example 2 • Other functions



Solution. $\lim_{x\to 0^+} \ln x = -\infty$; $\lim_{x\to \frac{\pi}{2}^-} \tan x = +\infty$, $\lim_{x\to \frac{\pi}{2}^+} \tan x = -\infty$.

Common mistakes

- Writing $\lim_{x\to c} f(x) = \infty$ when the two one-sided limits have opposite signs.
- Claiming every zero of a denominator gives an asymptote (check for cancellation first).

Quick check

1. $\lim_{x\to -3^+} \dfrac{2}{x+3}$ 2. $\lim_{x\to 1} \dfrac{-1}{(x-1)^2}$ 3. Vertical asymptotes of $\dfrac{x}{x^2-x}$?

Answers. 1. $+\infty$. 2. $-\infty$. 3. $\frac{x}{x(x-1)}$: only $x = 1$ (hole at $x = 0$).